No.

Date:

Tugas 5

DERIVATIF

1. Gunakan rumus yang ada untuk menentukan derivatif turunan dari fungsi :

a $y = 2x^3 + x^2 - 10x + 5$

b $f(x) = 2\sqrt{x} - \frac{1}{x} + \frac{3}{x^2}$

c $f(x) = e^x + \cos x$

d). $f(x) = \frac{1-x}{1+x}$, $x \neq -1$

e). $f(t) = \sin 2t - \cos 2t$

d]. $f(x) = x^2 \sin x$

2. sebuah bola diemparkan dimana posisi untuk setiap saat t (sec) memenuhi persamaan : $s(t) = 9t - t^2$ meter

a Hitung kecepatan rata rata antara $t = 2{,}00$ sec dan $t = 2{,}05$ dan hitung kecepatan saat pada $t = 2{,}00$ sec. Berapa perbedaan keduanya, artinya

b kapan bola mencapai tanah ($s = 0$) dan tinggi maksimum

3. muatan listrik memenuhi : $q(t) = \sin 2t$

Hitung arus rata rata antara $t = \frac{\pi}{4}$ dan $t = \frac{\pi}{2}$ dan hitung arus saat $t = \frac{\pi}{3}$

Jawaban

1. a $y = 2x^3 + x^2 - 10x + 5$

$y' = 6x^2 + 2x - 10$

b $f(x) = 2\sqrt{x} - \frac{1}{x} + \frac{3}{x^2} \longrightarrow f'(x) = \frac{1}{\sqrt{x}} + x^{-2} - 6x^{-3}$

$f'(x) = 2x^{\frac{1}{2}} - x^{-1} + 3x^{-2}$

$f'(x) = \frac{1}{2} \cdot 2 \cdot x^{\frac{1}{2}-1} + 1x^{-1-1} + (-6)x^{-2-1}$

$= x^{-\frac{1}{2}} x^{-2} + (-6)x^{-3}$

$= \frac{1}{\sqrt{x}} + \frac{1}{x^2} - \frac{6}{x^3}$

c $f(x) = e^x + \cos x$

$f'(x) = e^x - \sin x$

d $f(x) = x^2 \cdot \sin x$

$f'(x) = (x^2)' \sin x + x^2 (\sin x)'$

$= 2x \sin x + x^2 \cos x$

No.

Date:

d $f(x) = \frac{1-x}{1+x}$. $x \neq -1$

$f'(x) = \frac{u'v - uv'}{v^2} = \frac{(-1)(1+x) - 1(1-x)}{(1+x)^2} = \frac{-2}{(1+x)^2}$

e $f(t) = \sin 2t - \cos 2t$

$f'(t) = 2\cos 2t + 2\sin 2t$

$2(\cos 2t + \sin 2t)$

2. a $t_1 = 2{,}00$ sec $\rightarrow s(2{,}00) = 9(2{,}00) - (2{,}00)^2 = 14$

$t_2 = 2{,}05$ sec $\rightarrow s(2{,}05) = 9(2{,}05) - (2{,}05)^2 = 18{,}45 - 4{,}2025 = 14{,}2475$

$\therefore \vec{V} = \frac{\Delta s}{\Delta t} = \frac{s(t_2) - s(t_1)}{t_2 - t_1} = \frac{14{,}2475 - 14}{2{,}05 - 2{,}00} = \frac{0{,}2475}{0{,}05} = 4{,}95$ m/s

• kecepatan saat pada $t = 2{,}00$ sec

$s(t) = 9t - t^2$

$s'(t) = 9 - 2t$

$V(2{,}00) = 9 - 2.2{,}00 = 5$ m/s

perbedaan keduanya $5 - 4{,}95 = 0{,}05$ m/s

artinya kecepatan rata rata mendekati kecepatan saat $t = 2{,}00$ sec

3. persamaan $q(t) = \sin(2t)$

$I = (t) = \frac{dq}{dt}$

$I(t) = q'(t)$

$= 2\cos(2t)$

• saat $t = \frac{\pi}{4}$

$q\left(\frac{\pi}{4}\right) = \sin\left(2 \cdot \frac{\pi}{4}\right)$

$= \sin(90^\circ) = 1$

• saat $t = \frac{\pi}{2}$

$q\left(\frac{\pi}{2}\right) = \sin\left(2 \cdot \frac{\pi}{2}\right)$

$= \sin(180^\circ) = 0$

Perubahan muatan (Δq)

$\Delta q = 0 - 1 = -1$

selang waktu (Δt)

$\Delta t = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

• I rata rata $= \frac{\Delta q}{\Delta t}$

$= \frac{-1}{\frac{\pi}{4}} = \frac{-4}{\pi}$

• arus sesaat $t = \frac{\pi}{3}$

$I\left(\frac{\pi}{3}\right) = 2\cos\left(2 \cdot \frac{\pi}{3}\right)$

$= 2\cos(120)$

$= 2\left(-\frac{1}{2}\right)$

$= -1$

arus sesaat pada $t = \frac{\pi}{3}$ adalah -1 satuan arus

No.

Date:

2. b kapan bola mencapai tanah ($s = 0$)

$s(t) = 9t - t^2$

$0 = 9t - t^2$

$9t - t^2 = 0$

$t(-t + 9) = 0$ $\quad (t - 9)$ $\quad (t + 9) = 0$

$t = 0$ $\qquad t = 9$

$s(t) = 9t - t^2$

$s'(t) = 9 - 2t$

$0 = 9 - 2t$

$2t = 9$

$t = \frac{9}{2} = 4{,}5\ s$

$q(t) = \sin t \qquad t_1 = \frac{\pi}{4} \quad t_2 = \frac{\pi}{2}$

hitung rata rata arus dan arus sesaat pada $t = \frac{\pi}{4}$

- rata rata arus